Hallow App Architecture Demo

A Multi-Database Approach

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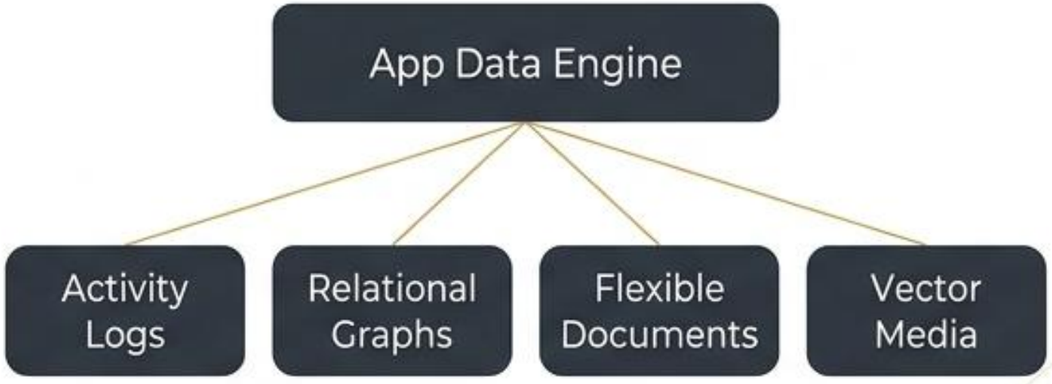
The Vision

A functional database replica of the Hallow App. Designed exclusively for registered user interactions, managing complex social features, personal notes, and integrated AI chatbot conversations.

The Technical Goal



Implement a polyglot persistence architecture. Instead of forcing one database to do everything, the system routes distinct data formats to highly specialized database engines for optimal scale and performance.

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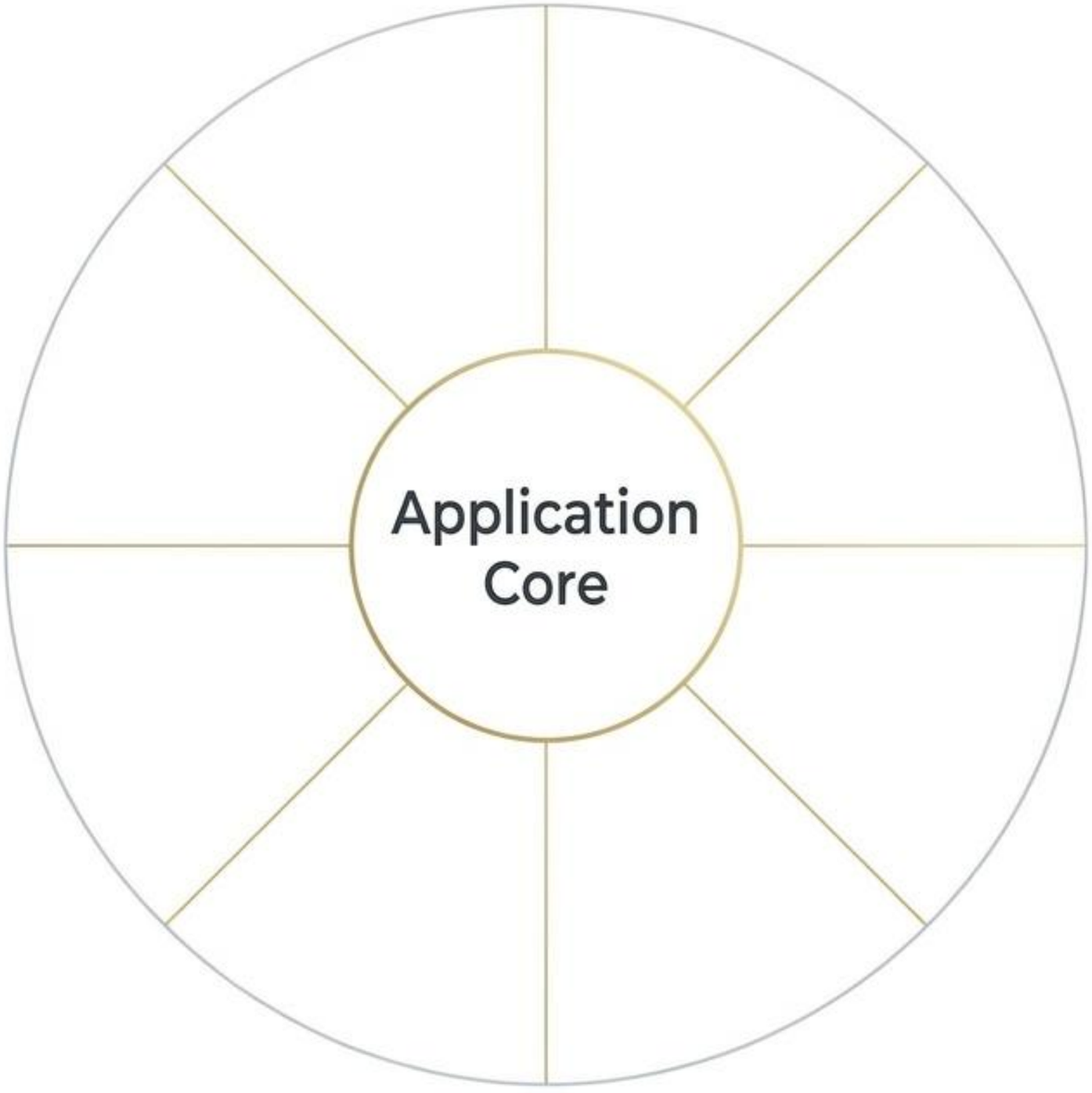
Cassandra

High-Volume Activity Logs (The Memory)

Dgraph

Relational Networking (The Social Fabric)



MongoDB

Flexible Core Data (The Core)

ChromaDB

Semantic AI Engine (The Mind)

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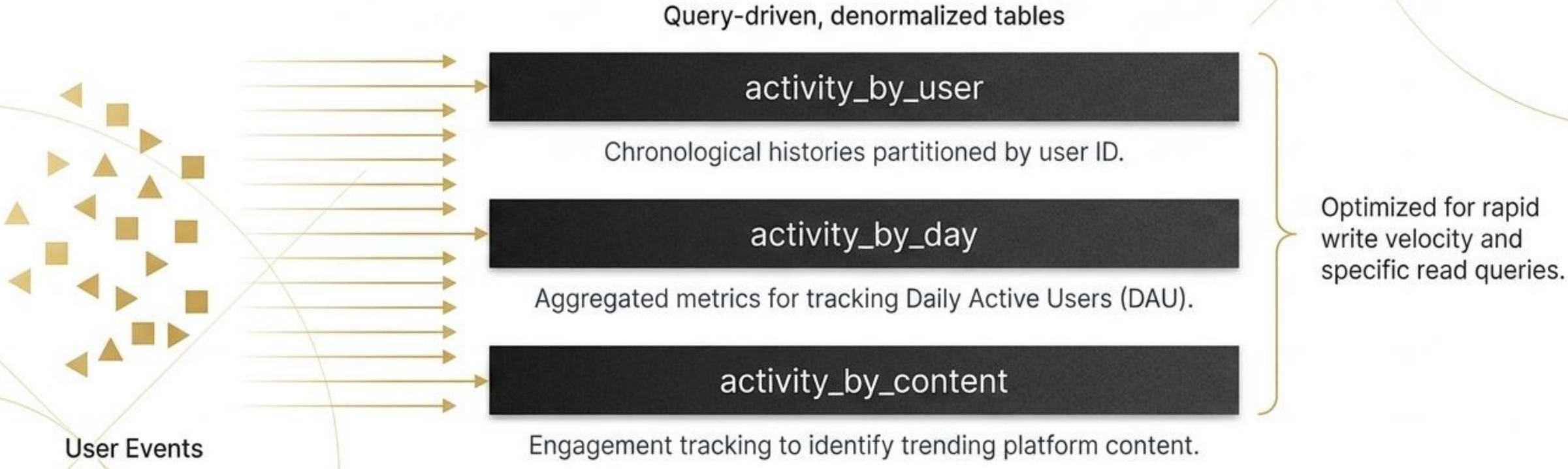
Cassandra: The Log Engine

Captures high-throughput user activity and session tracking at scale without relying on expensive database joins.



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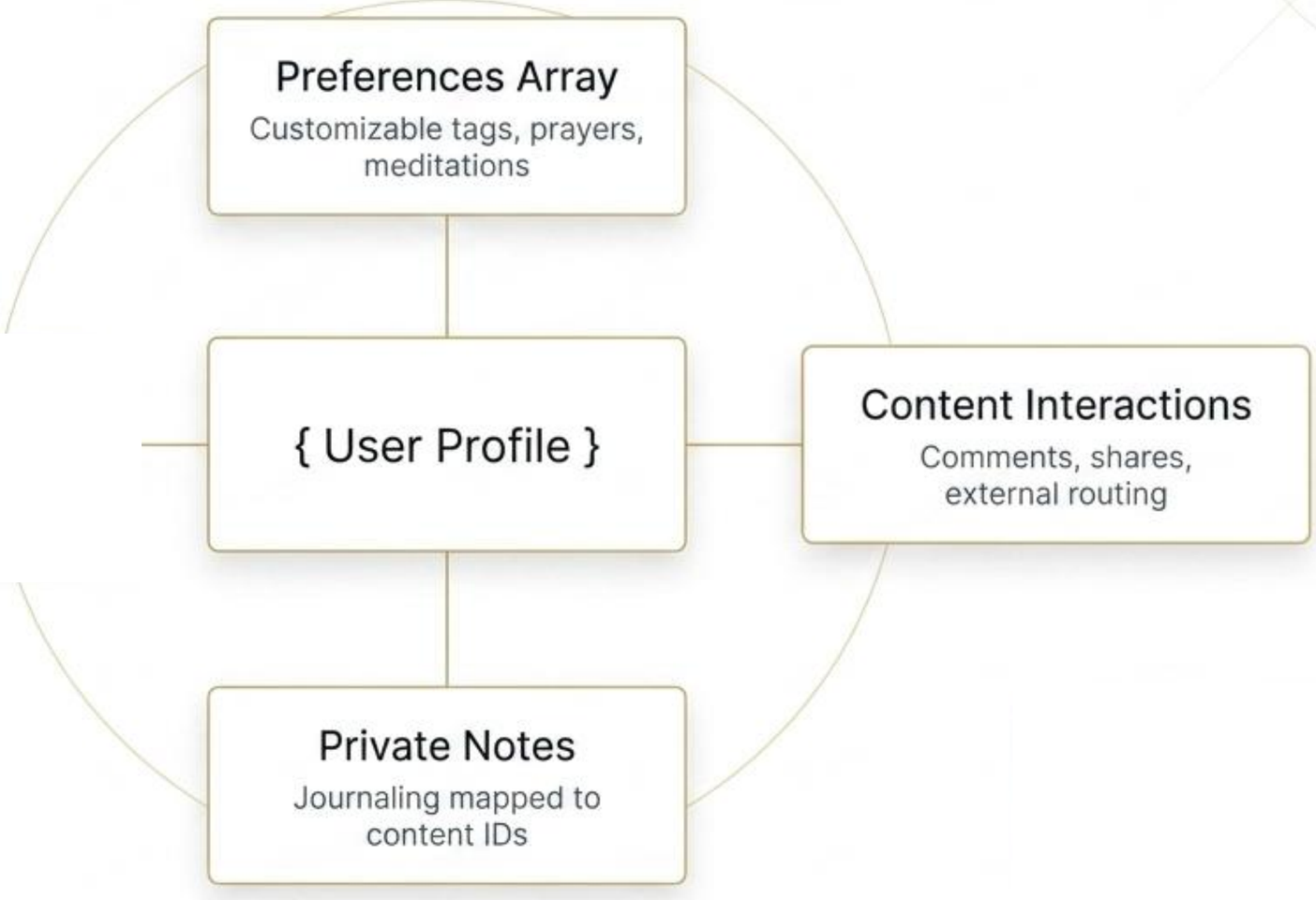
MongoDB: The Core System

Manages flexible, user-centric data that evolves dynamically without rigid schema constraints.



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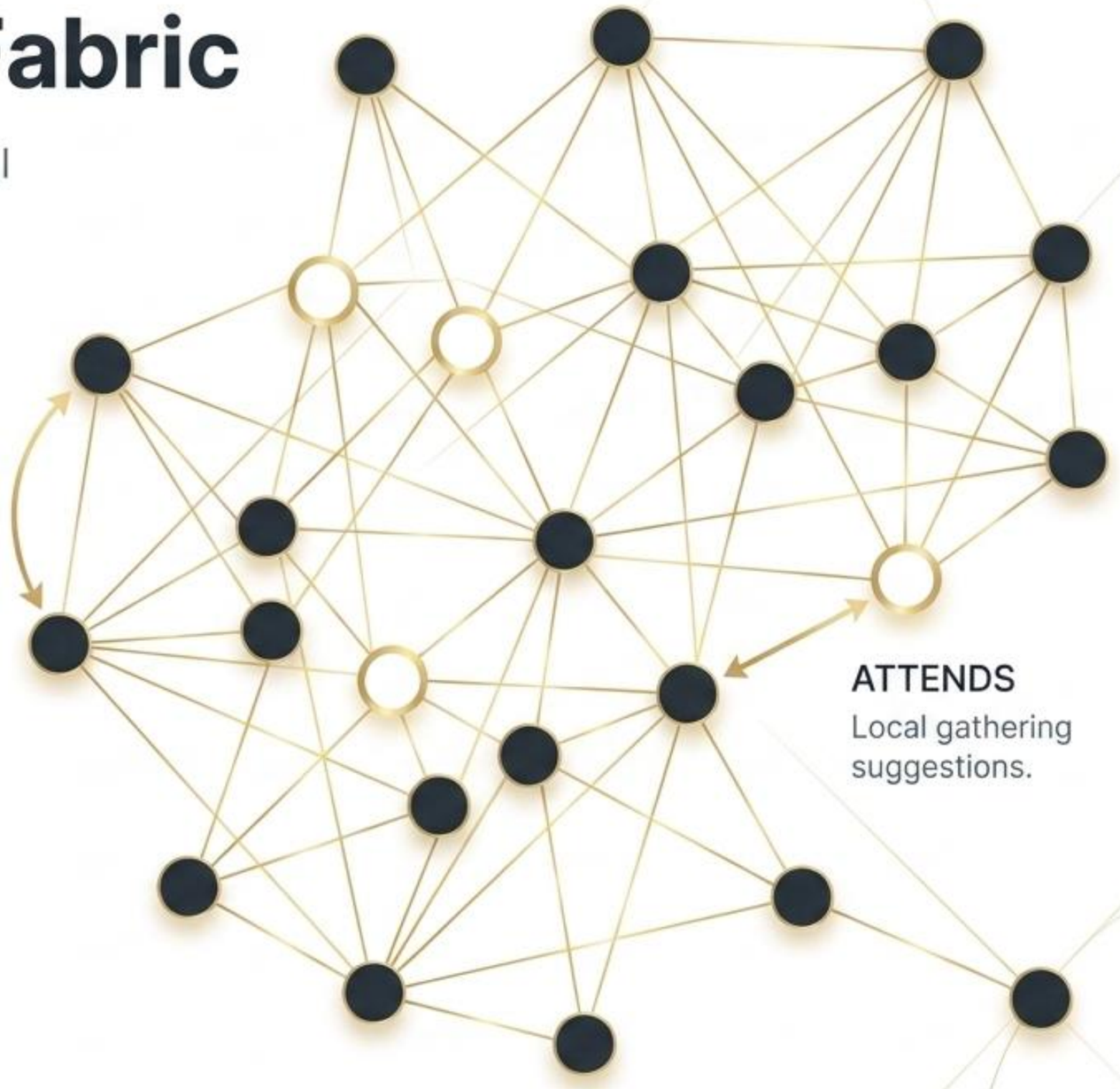
Dgraph: The Social Fabric

Maps complex social connections and geographical relationships for rapid, multi-level traversal.



The Graph Advantage

Unlocks rapid community discovery and geospatial event filtering that relational databases cannot efficiently compute. Highly connected data traversals are executed instantly.

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ChromaDB - Vector Store

- Embeddings- text turned into dense vectors
- Semantic retrieval - nearest-neighbor similarity
- RAG contest – relevant docs surfaced for generation

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Demonstration

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Lessons Learned

Data Fit

A single database cannot optimize for flexible documents, heavy writes, complex relationships, and vector math simultaneously. Polyglot persistence is essential for modern platforms.

Denormalization Wins

For immense scale, designing database tables around specific application queries (Cassandra) is vastly superior to normalizing data and relying on compute-heavy joins.

Meaning Over Matching

Implementing ChromaDB revealed that semantic search (vectors) fundamentally changes user interaction, surfacing relevant content by underlying intent rather than rigid keyword matching.

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Thank you!

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